

13.2 - The Golden Ratio

Problem Solving, Math 1000

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1 Definition

The **Golden Ratio**, often denoted by the Greek letter ϕ , is an irrational number (which loosely means it has a nonrepeating, infinite decimal expansion) with the **golden property**: $\phi^2 = \phi + 1$. Substituting x for ϕ and manipulating the equation, we may employ the quadratic formula to see:

$$\begin{aligned}x^2 &= x + 1 \\x^2 - x - 1 &= 0 \\x &= \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-1)}}{2(1)} \\x &= \frac{1 \pm \sqrt{5}}{2}\end{aligned}$$

The positive solution found above is what we call ϕ , the golden ratio. That is to say,

$$\phi = \frac{1 + \sqrt{5}}{2}$$

In decimal form, this is

$$\phi \approx 1.6180339887$$

The golden ratio has a variety of nice properties we will briefly look into.

2 Theorem (The Golden Power Formula)

The golden ratio, ϕ , is deeply tied to the Fibonacci numbers. One such way this expresses itself is by the following formula, which is called the golden power formula:

$$\phi^N = F_N \phi + F_{N-1}$$

3 Observe

The connection between the golden ratio and the fibonacci numbers can be even more clearly seen by looking at the sequence of numbers obtained by dividing a Fibonacci number by the proceeding Fibonacci number (i.e. $\frac{F_N}{F_{N-1}}$). In fraction form, this gives:

$$\frac{1}{1}, \frac{2}{1}, \frac{3}{2}, \frac{5}{3}, \frac{8}{5}, \frac{13}{8}, \frac{21}{13}, \frac{34}{21}, \dots$$

The connection to the golden ratio is not particularly obvious when we write these in fraction form. However, writing these fractions as decimals (rounding to six decimal places for convenience), it becomes much clearer:

$$1, 2, 1.5, 1.666667, 1.6, 1.625, 1.615385, 1.619048, \dots$$

In fact, it is the case that the ratios of successive Fibonacci numbers converges to the golden ratio (that is, as N gets very large, $\frac{F_N}{F_{N-1}} \approx \phi$).